```
(base) mac@Chais-MacBook-Air ~ % aws ec2 describe-launch-template-versions \
  --launch-template-name itmo-444-544-uploader-lt \
  --query 'LaunchTemplateVersions[*].[VersionNumber,CreateTime]' \
  --output table \
[  --region us-east-2
-----------------------------------
|  DescribeLaunchTemplateVersions  |
+---+------------------------------+
|  2|  2026-10-09T21:14:00+00:00   |
|  1|  2026-10-09T20:51:36+00:00   |
+---+------------------------------+
```

```
Last login: Fri Oct  9 15:04:43 on ttys000
[(base) mac@Chais-MacBook-Air ~ % BUCKET_NAME="itmo-444-544-ckirankumar-fileshare"
[(base) mac@Chais-MacBook-Air ~ % echo "$BUCKET_NAME"
itmo-444-544-ckirankumar-fileshare
(base) mac@Chais-MacBook-Air ~ % aws s3api create-bucket \
  --bucket "$BUCKET_NAME" \
  --region us-east-2 \
[  --create-bucket-configuration LocationConstraint=us-east-2
{
    "Location": "http://itmo-444-544-ckirankumar-fileshare.s3.amazonaws.com/",
    "BucketArn": "arn:aws:s3:::itmo-444-544-ckirankumar-fileshare"
}
[(base) mac@Chais-MacBook-Air ~ % aws s3 ls | grep fileshare
2026-10-06 23:28:49 itmo-444-544-895207965179-fileshare
2026-10-09 15:09:42 itmo-444-544-ckirankumar-fileshare
(base) mac@Chais-MacBook-Air ~ % aws s3api delete-bucket \
  --bucket itmo-444-544-895207965179-fileshare \
[  --region us-east-2
(base) mac@Chais-MacBook-Air ~ % aws s3api delete-bucket \
  --bucket itmo-444-544-ckirankumar-fileshare \
[  --region us-east-2
(base) mac@Chais-MacBook-Air ~ %
```

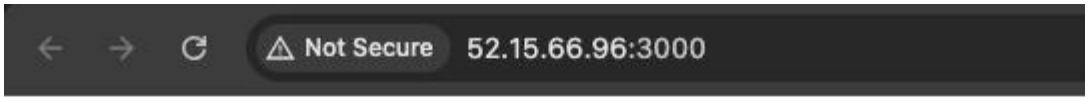
```
(base) mac@Chais-MacBook-Air ~ % aws iam put-role-policy \
  --role-name itmo-444-544-uploader-role \
  --policy-name uploader-put-only \
[  --policy-document file://uploader-policy.json
(base) mac@Chais-MacBook-Air ~ % aws iam get-role-policy \
  --role-name itmo-444-544-uploader-role \
[  --policy-name uploader-put-only
{
    "RoleName": "itmo-444-544-uploader-role",
    "PolicyName": "uploader-put-only",
    "PolicyDocument": {
        "Version": "2012-10-17",
        "Statement": [
            {
                "Effect": "Allow",
                "Action": [
                    "s3:PutObject"
                ],
                "Resource": "arn:aws:s3:::itmo-444-544-ckirankumar-fileshare/shared.txt"
            }
        ]
    }
}
(base) mac@Chais-MacBook-Air ~ %
```

```
(base) mac@Chais-MacBook-Air ~ % aws iam get-instance-profile \
  --instance-profile-name itmo-444-544-uploader-profile
{
    "InstanceProfile": {
        "Path": "/",
        "InstanceProfileName": "itmo-444-544-uploader-profile",
        "InstanceProfileId": "AIPA5A3UNQH5SSYURFNB3",
        "Arn": "arn:aws:iam::895207965179:instance-profile/itmo-444-544-uploader-profile",
        "CreateDate": "2026-10-09T20:19:56+00:00",
        "Roles": [
            {
                "Path": "/",
                "RoleName": "itmo-444-544-uploader-role",
                "RoleId": "AROA5A3UNQH56O4ZCJ5ZV",
                "Arn": "arn:aws:iam::895207965179:role/itmo-444-544-uploader-role",
                "CreateDate": "2026-10-07T04:34:46+00:00",
                "AssumeRolePolicyDocument": {
                    "Version": "2012-10-17",
                    "Statement": [
                        {
                            "Effect": "Allow",
                            "Principal": {
                                "Service": "ec2.amazonaws.com"
                            },
                            "Action": "sts:AssumeRole"
                        }
                    ]
                }
            }
        ],
        "Tags": []
    }
}
```

```
(base) mac@Chais-MacBook-Air lab03 % find ~/lab03 -type f
/Users/mac/lab03/viewer-app/package.json
/Users/mac/lab03/viewer-app/app.js
/Users/mac/lab03/uploader-app/package.json
/Users/mac/lab03/uploader-app/app.js
```

```
(base) mac@Chais-MacBook-Air ~ % ./create_app_stack.sh
== S3 bucket ==
{
    "Location": "http://itmo-444-544-ckirankumar-fileshare.s3.amazonaws.com/",
    "BucketArn": "arn:aws:s3:::itmo-444-544-ckirankumar-fileshare"
}
== Uploader IAM role ==
== Viewer IAM role ==
{
    "Role": {
        "Path": "/",
        "RoleName": "itmo-444-544-viewer-role",
        "RoleId": "AROA5A3UNQH5QZTCAAM5E",
        "Arn": "arn:aws:iam::895207965179:role/itmo-444-544-viewer-role",
        "CreateDate": "2026-10-09T20:51:12+00:00",
        "AssumeRolePolicyDocument": {
            "Version": "2012-10-17",
            "Statement": [
                {
                    "Effect": "Allow",
                    "Principal": {
                        "Service": "ec2.amazonaws.com"
                    },
                    "Action": "sts:AssumeRole"
                }
            ]
        }
    }
}
{
    "InstanceProfile": {
        "Path": "/",
        "InstanceProfileName": "itmo-444-544-viewer-profile",
        "InstanceProfileId": "AIPA5A3UNQH5YCPOLWRPI",
        "Arn": "arn:aws:iam::895207965179:instance-profile/itmo-444-544-viewer-profile",
        "CreateDate": "2026-10-09T20:51:16+00:00",
        "Roles": []
    }
}
Waiting 15 seconds for IAM propagation...
== Building launch templates ==
== Launching instances ==
Waiting for both instances to run...
-------------------------------------------------------------------
|                       DescribeInstances                         |
+---------------------+----------+--------------+-----------------+
|  i-0b7b9f9b56613b2dd |  running |  viewer-app   |  52.15.66.96    |
|  i-07b3d6a67fe8fbb4c |  running |  uploader-app |  18.222.240.72  |
+---------------------+----------+--------------+-----------------+
Allow a few minutes for app installation before opening port 3000.
(base) mac@Chais-MacBook-Air ~ %
```

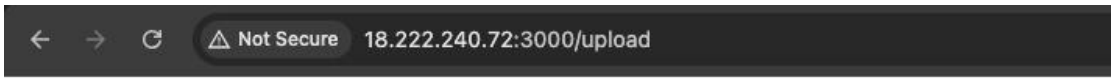
```
(base) mac@Chais-MacBook-Air ~ % aws ec2 describe-launch-templates \
  --launch-template-names itmo-444-544-uploader-lt itmo-444-544-viewer-lt \
  --query 'LaunchTemplates[*].[LaunchTemplateName,LaunchTemplateId,LatestVersionNumber]' \
  --output table \
  --region us-east-2
-------------------------------------------------------------
|                DescribeLaunchTemplates                    |
+--------------------------+-----------------------+----+
|  itmo-444-544-viewer-lt  |  lt-0c9f444231995459a |  1 |
|  itmo-444-544-uploader-lt |  lt-0f237a79e9be71bb8 |  1 |
+--------------------------+-----------------------+----+
(base) mac@Chais-MacBook-Air ~ % aws iam get-role-policy \
  --role-name itmo-444-544-viewer-role \
  --policy-name itmo-444-544-viewer-role-policy \
  --no-cli-pager
{
    "RoleName": "itmo-444-544-viewer-role",
    "PolicyName": "itmo-444-544-viewer-role-policy",
    "PolicyDocument": {
        "Version": "2012-10-17",
        "Statement": [
            {
                "Effect": "Allow",
                "Action": [
                    "s3:GetObject"
                ],
                "Resource": "arn:aws:s3:::itmo-444-544-ckirankumar-fileshare/shared.txt"
            },
            {
                "Effect": "Allow",
                "Action": [
                    "s3:ListBucket"
                ],
                "Resource": "arn:aws:s3:::itmo-444-544-ckirankumar-fileshare"
            }
        ]
    }
}
(base) mac@Chais-MacBook-Air ~ %
```



Viewer

Hello from my first upload!

Refresh



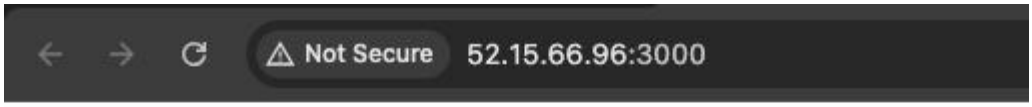
Upload successful.

Upload another



Upload successful.

Upload another



Viewer

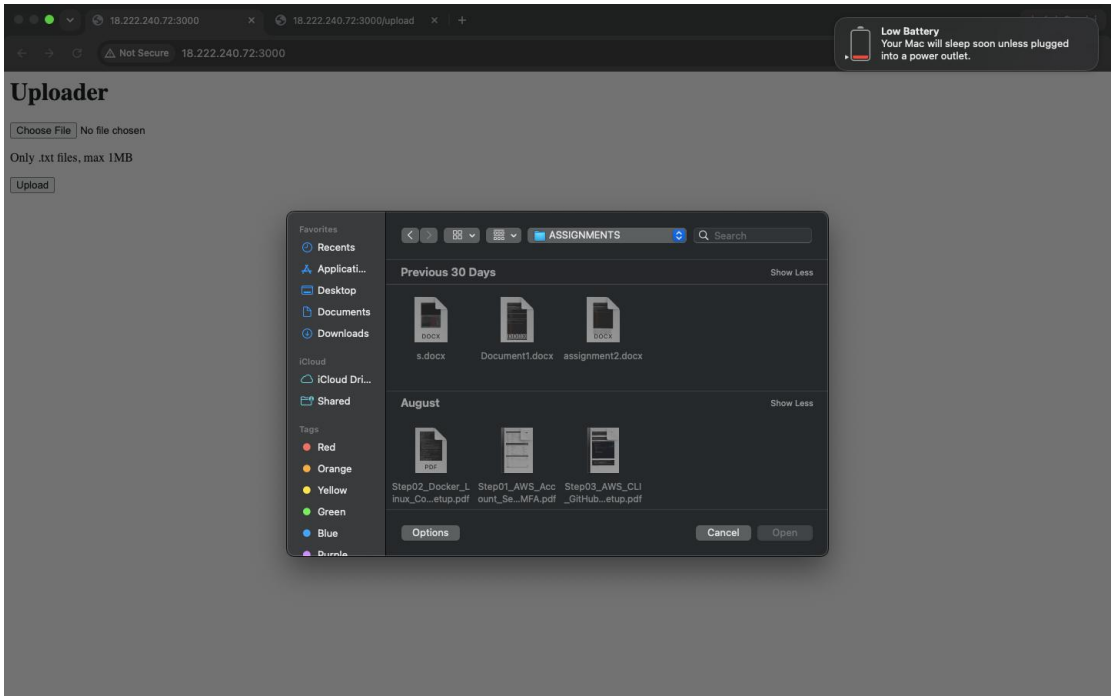
```
This is my second upload. The content has changed!
```

Refresh



Upload failed: file exceeds the 1MB limit.

Try again




```
(base) mac@Chais-MacBook-Air ~ % aws ec2 describe-launch-template-versions \
  --launch-template-name itmo-444-544-uploader-lt \
  --query 'LaunchTemplateVersions[*].[VersionNumber,CreateTime]' \
  --output table \
[  --region us-east-2
-------------------------------------
|  DescribeLaunchTemplateVersions   |
+---+-------------------------------+
|  2|  2026-10-09T21:14:00+00:00    |
|  1|  2026-10-09T20:51:36+00:00    |
+---+-------------------------------+
```



```
(base) mac@Chais-MacBook-Air ~ % aws ec2 describe-launch-template-versions \
  --launch-template-name itmo-444-544-uploader-lt \
  --query 'LaunchTemplateVersions[*].[VersionNumber,CreateTime]' \
  --output table \
  --region us-east-2
----------------------------------
|  DescribeLaunchTemplateVersions  |
+---+------------------------------+
|  2|  2026-10-09T21:14:00+00:00   |
|  1|  2026-10-09T20:51:36+00:00   |
+---+------------------------------+
```


```
(base) mac@Chais-MacBook-Air ~ % aws ec2 describe-launch-template-versions \
  --launch-template-name itmo-444-544-uploader-lt \
  --query 'LaunchTemplateVersions[*].[VersionNumber,CreateTime]' \
  --output table \
  --region us-east-2
----------------------------------
|  DescribeLaunchTemplateVersions  |
+---+------------------------------+
|  2|  2026-10-09T21:14:00+00:00   |
|  1|  2026-10-09T20:51:36+00:00   |
+---+------------------------------+
```



```
+---+------------------------------+
(base) mac@Chais-MacBook-Air ~ % cd ~
./delete_app_stack.sh
Terminating i-07ba90bfb2f346ae9
Terminating i-0bf211f3a104b98b3
== Deleting launch templates ==
{
    "LaunchTemplate": {
        "LaunchTemplateId": "lt-0f237a79e9be71bb8",
        "LaunchTemplateName": "itmo-444-544-uploader-lt",
        "CreateTime": "2026-10-09T20:51:36+00:00",
        "CreatedBy": "arn:aws:iam::895207965179:user/chaya",
        "DefaultVersionNumber": 1,
        "LatestVersionNumber": 2,
        "Operator": {
            "Managed": false
        }
    }
}
{
    "LaunchTemplate": {
        "LaunchTemplateId": "lt-0c9f444231995459a",
        "LaunchTemplateName": "itmo-444-544-viewer-lt",
        "CreateTime": "2026-10-09T20:51:39+00:00",
        "CreatedBy": "arn:aws:iam::895207965179:user/chaya",
        "DefaultVersionNumber": 1,
        "LatestVersionNumber": 2,
        "Operator": {
            "Managed": false
        }
    }
}
== Removing lab profiles and roles ==
== Emptying and deleting the lab bucket ==
{
    "BucketArn": "arn:aws:s3:::itmo-444-544-ckirankumar-fileshare",
    "BucketRegion": "us-east-2",
    "AccessPointAlias": false
}
delete: s3://itmo-444-544-ckirankumar-fileshare/shared.txt
Cleanup complete.
(base) mac@Chais-MacBook-Air ~ %
```

The Uploader and Viewer use separate IAM roles to follow the principle of least privilege. The Uploader needs only s3:PutObject permission to write shared.txt, while the Viewer receives s3:GetObject permission to read that file and s3:ListBucket permission on the bucket. Separate roles prevent the Viewer from modifying the file and the Uploader from reading it, limiting what each service can do if it is compromised.

The app code is embedded directly in the EC2 user-data script and recreated on the instance during startup. Therefore, the instance does not need Git or S3 credentials to download its application code; it uses its IAM role only for the app's S3 operations.